claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a45d7bb7db7e17b9f.jsonl`
- SHA-256: `b40b899f229022d2793ca4c328b8f52c54b776d777790dd3ed15c4311691a834`
- Source modified: `2026-06-16T03:20:30+00:00`
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- project: `wf_3d536535-241`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`

Transcript

1. user (2026-06-16T03:19:49.183Z)

Source Extractor

```
Research question: "Research question: **What concrete, recent (2022?2025) techniques and
evaluation practices should a local, commercially-licensed badminton rally-detector adopt to
close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16
golden videos)?** Produce an adversarially-verified, cited report organized around the four
dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over
generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):
- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a
heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff"
confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits
far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it
does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active"
iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold;
active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary
split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves +
knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-
frame visibility).
- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom
licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM
(Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference;
minors excluded from any training pool; runs on a single local GPU.
- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture
skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) +
late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet,
See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation,
tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-
one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-
count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE,
a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment
leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal
windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-
window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ?
what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented
actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression
and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression),
BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads),
smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,
```

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RLOOCV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:
URL: https://yiskw713.github.io/asrf/
Title: ASRF: Alleviating Over-segmentation Errors by Detecting Action Boundaries (Ishikawa et al., WACV 2021)
Found via: TAL boundary / over-segmentation methods (academic, recent) search

Task
1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
   - be a concrete, checkable statement (not vague generalities)
   - include a direct quote from the source as support
   - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:19:51.974Z)

I'll fetch the ASRF paper page and extract relevant claims.

3. user (2026-06-16T03:20:00.560Z)

ASRF Technical Claims Summary

Problem Addressed
The paper tackles over-segmentation errors in temporal action segmentation. The framework aims to refine frame-level action classification using predicted action boundaries.

Technical Approach

ASRF employs a two-branch architecture: "(i) an Action Segmentation Branch (ASB) and the Boundary Regression Branch (BRB)." The ASB performs frame-wise classification while the BRB "regresses the action boundary probabilities." These boundary predictions then "refine the output from the ASB."

Key Contributions

The authors propose: "(i) a framework for temporal action segmentation...which divides temporal action segmentation into frame-wise action classification and action boundary regression." Additionally, they introduce "(ii) a loss function for smoothing the transition of action probabilities."

Quantitative Results

Performance improvements are substantial: "up to 13.7% in terms of segmental edit distance and up to 16.1% in terms of segmental F1 score" over prior methods. Datasets evaluated include "50 Salads, GTEA and Breakfast dataset."

Publication Details

Published in **WACV 2021** by Yuchi Ishikawa, Seito Kasai, Yoshimitsu Aoki, and Hirokatsu Kataoka. Code is available on [GitHub](https://github.com/yiskw713/asrf).

4. assistant (2026-06-16T03:20:05.399Z)

I have the content. Let me also verify the GitHub repo for licensing since the research question has hard license constraints.

5. user (2026-06-16T03:20:11.957Z)

License and Framework Information

License: This repository is released under the MIT License, as stated in the repository documentation.

Framework Method: The implementation focuses on alleviating over-segmentation errors through action boundary detection. The approach builds upon temporal convolutional networks, specifically referencing MS-TCN architecture from prior work.

Key References:
The code builds on two foundational approaches:
- "Temporal Convolutional Networks for Action Segmentation and Detection" (CVPR 2017)
- "MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation" (CVPR 2019)

Note on Metrics: The provided content does not include specific performance metrics or quantitative results. The repository structure suggests experimental results are stored in the `result` directory, but actual benchmark numbers are not presented in this page content.

6. assistant (2026-06-16T03:20:15.542Z)

Both fetches succeeded. The page is the official project page for a peer-reviewed WACV 2021 paper (primary research), and the reference implementation is MIT-licensed ? directly relevant to the license-clean, over-segmentation-control constraints in the research question.

7. user (2026-06-16T03:20:25.163Z)

Structured output provided successfully

8. assistant (2026-06-16T03:20:30.398Z)

Extracted 5 falsifiable claims from the ASRF project page (primary source: official WACV 2021 paper page, plus the MIT-licensed reference repo at https://github.com/yiskw713/asrf). The source is highly relevant to Dimension 1 (windowing/boundaries, over-segmentation control): ASRF's boundary-regression branch + probability-smoothing loss directly target the indexer's "fails to segment / merges into mega-windows" failure, it is MIT-licensed (license-clean), and it reports concrete gains (up to +16.1% segmental F1, +13.7% edit distance over MS-TCN-style baselines on 50 Salads/GTEA/Breakfast).